Chapter 8: Playing with Constructions

Worksheet 8: Geometric Constructions

Learning Objective: Students will use a compass and ruler to construct: perpendicular bisectors of line segments, angle bisectors, angles of 60 and 90, equilateral triangles, and circles; understand the properties underlying each construction.

Worked Example

To construct a perpendicular bisector of a line segment AB:

1. Open compass to more than half of AB.
2. Draw arcs above and below AB from both A and B.
3. Join the two intersection points. This line bisects AB at right angles.

Questions 1–5: Easy / Recall

Q1. Name two instruments used in geometric constructions.

Q2. What does a perpendicular bisector do to a line segment?

Q3. An angle bisector divides an angle into __________.

Q4. Construct a line segment of length 6.5 cm. (Teacher Verification Required)

Q5. What is the angle in an equilateral triangle? How do we construct 60?

Questions 6–10: Basic Application

Q6. Construct a perpendicular bisector of a line segment 8 cm long. (Teacher Verification Required)

Q7. Construct an angle of 60. (Teacher Verification Required)

Q8. Using your 60 construction, construct 30. Describe the steps. (Teacher Verification Required)

Q9. Construct a 90 angle using only a compass and ruler. (Teacher Verification Required)

Q10. Draw a circle of radius 4 cm. Mark its centre, a diameter, and two radii. (Teacher Verification Required)

Questions 11–15: Practice and Skill Building

Q11. Construct an equilateral triangle with side 5 cm. Measure all three angles. (Teacher Verification Required)

Q12. Construct the angle bisector of a 120 angle. What angle does each half measure? (Teacher Verification Required)

Q13. Construct $\angle ABC = 75$ using constructions of 60 and 15. Describe your method. (Teacher Verification Required)

Q14. Draw a line PQ. Construct a perpendicular to PQ at a point M on the line. (Teacher Verification Required)

Q15. Construct an isosceles triangle with equal sides 6 cm and base 4 cm. (Teacher Verification Required)

Questions 16–18: Higher Order Thinking

Q16. Using only compass and ruler, is it possible to construct exactly 45? Describe the steps. (Teacher Verification Required)

Q17. The perpendicular bisectors of all three sides of a triangle meet at a single point. What is this point called? What is its special property? (No construction required – explain in words.)

Q18. Construct a regular hexagon inside a circle of radius 5 cm. Explain why all sides of the hexagon equal the radius. (Teacher Verification Required)

Questions 19–20: Real-Life Applications

Q19. A carpenter wants to mark the exact centre of a rectangular wooden board. Describe a construction method using only a ruler (no measuring of the board's dimensions is allowed; only drawing diagonals).

Q20. An engineer wants to build a perfectly square room. She has a long rope. Explain how she could use the rope and the concept of a perpendicular bisector to mark right-angle corners for the room.

Reflection Question: Why are compass-and-ruler constructions preferred over measurement in geometry? What makes them more precise?